A Digital Logic-Based Approach on mobile robot Obstacle Avoidance Using IR Sensors and RTL NOT Gate

1nd Reshad Ul Karim
Department of Computer Science and Engineering
BRAC University
Dhaka, Bangladesh
reshad.ul.karim@g.bracu.ac.bd

2st Abrar Samin
Department of Computer Science and Engineering
BRAC University
Dhaka, Bangladesh
abrar.samin@g.bracu.ac.bd

3rd Sammam Mahdi
Department of Computer Science and Engineering
BRAC University
Dhaka, Bangladesh
sammam.mahdi@g.bracu.ac.bd

4th Nashita Anjum
Department of Computer Science and Engineering
BRAC University
Dhaka, Bangladesh
nashita.anjum@g.bracu.ac.bd

***Abstract*—This paper presents the design and implementation of a microcontroller-free obstacle-avoiding robot using infrared (IR) sensors and discrete transistor-based logic gates. The proposed system demonstrates a foundational approach to autonomous navigation through the use of Resistor-Transistor Logic (RTL) for real-time decision-making. Obstacle detection is achieved via Infrared sensors equipped with operational amplifier(op-amp) based comparators, which convert analog reflectivity data into binary signals. These outputs are processed through RTL NOT gates implemented using NPN BJTs to generate motor control logic. The robot employs an L298N H-bridge motor driver to actuate two DC motors based on the inverted sensor signals. The system architecture eliminates the need for programmable components, offering a pedagogical model for beginners in robotics and digital electronics. Experimental validation demonstrates reliable obstacle detection and dynamic response under various obstacle scenarios, highlighting the practicality of low-cost, logic-driven autonomous systems.**

I. Introduction

Obstacle-avoiding robots are among the most fundamental robotics projects, especially for beginners learning embedded systems and digital logic. This robot uses infrared (IR) sensors for obstacle detection and applies transistor-based NOT gates to generate the logic required to control the direction of two motors. The system's real-time responsiveness and simplicity make it a practical and instructive robotics prototype.

II. Implementation

A. Components and Hardware Architecture

The obstacle-avoidance robotic platform was constructed using discrete components to demonstrate logic-level hardware control without the use of microcontrollers. The hardware components utilized are listed below:

- Infrared (IR) Obstacle Detection Sensors (×2)
- NPN Bipolar Junction Transistors (BJTs) (×2)
- Resistors: 15k Ω (×2), 100k Ω (×2), and 2.2k Ω (×2)
- L298N Dual H-Bridge Motor Driver Module (×1)
- DC Geared Motors with Wheels (×2)
- Two-Wheel Drive (2WD) Robot Chassis
- Power Supply: Dual 3.7V Li-ion batteries in series (Total: 7.4V)

This hardware configuration supports analog signal processing and logic inversion using RTL (Resistor-Transistor Logic), enabling a pedagogical approach to obstacle avoidance without programmable control units.

B. Infrared Obstacle Detection and Configuration

The IR sensors are integral to the robot's ability to detect obstacles in its forward path. Each IR module comprises an infrared LED (transmitter) and a photodiode or phototransistor (receiver), which work together to detect reflected infrared light from nearby objects [1] . When no obstacle is present in front of the sensor, minimal IR light is reflected back to the receiver, and the output remains at a digital high voltage level (approximately 3.3 V). However, when an object is within detection range, the reflected IR light activates the receiver, and the output voltage drops significantly, typically to around 400 mV.

Internally, the sensor employs an op-amp-based comparator circuit that processes the analog signal from the receiver and converts it into a digital output. This comparator compares the received IR intensity against a predefined threshold voltage. If the received intensity exceeds the threshold (indicating an obstacle), the comparator output switches to LOW. The use of a comparator ensures noise immunity and sharp digital transitions, which are critical for reliable real-time obstacle detection. This design is visualized in the schematic diagram shown in Fig. 5.

The IR module also features an onboard variable resistor (potentiometer), labeled as the distance adjustment knob. By rotating this potentiometer, the threshold level of the comparator can be fine-tuned, effectively adjusting the sensor's detection range. This range adjustment is dependent on the supply voltage provided at the VCC pin and the reflective characteristics of the obstacle surface. The potentiometer's location and layout are illustrated in the module's pin configuration diagram in Fig. 1.

The output from the IR sensor is then passed to a transistor-based NOT gate circuit that performs logic inversion, ensuring that a low output (indicating an obstacle) becomes a logic HIGH input for the motor driver. This inversion stage is implemented using a resistor-transistor logic (RTL) configuration with an NPN BJT.

The standard pin configuration of the IR sensor module is shown in Fig. 1, typically comprising three terminals: VCC, GND, and digital output (OUT). The digital OUT pin is responsible for delivering the binary obstacle detection signal, which is processed further for motion control. The onboard potentiometer allows adjustment of the detection range based on VCC.

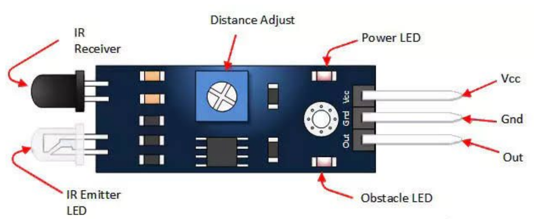

Fig. 1: Pin configuration of the IR obstacle sensor used in the design.

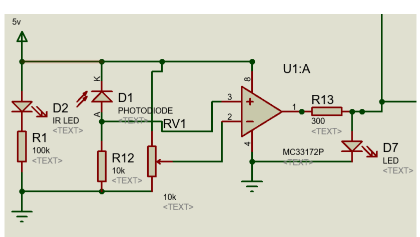

Fig. 2: Schematic of the IR sensor module showing internal op-amp-based comparator and external BJT-based NOT gate logic.

This configuration enables precise real-time detection and interpretation of environmental conditions, laying the foundation for an effective and responsive hardware-based navigation system.

C. RTL-Based NOT Gate Implementation Using BJT

To invert the IR sensor's output signal, a NOT gate is realized using an RTL configuration comprising an NPN BJT and resistors. When the IR sensor output is low (obstacle detected), the transistor remains in the cutoff region, and the collector voltage remains high. Conversely, when the IR output is high (no obstacle), the transistor saturates, pulling the output voltage to ground. This simple yet effective inversion mechanism is critical for interfacing with the motor driver module (refer to Fig. 5).

This hardware implementation offers students insight into digital logic gate behavior and transistor operation, bridging the gap between analog and digital electronics in robotics.

D. Motor Control via L298N Dual H-Bridge Driver

The L298N motor driver is employed to actuate two DC motors based on the logic signals obtained from the RTL NOT gates. It supports bidirectional control, allowing for both forward and reverse motion by toggling the logic input pins (IN1, IN2, IN3, IN4) [2]. The pinout of the L298N motor driver module used for the project is shown in Fig. 3. The enable pins (ENA and ENB) are tied to VCC to allow continuous operation, and PWM was not implemented as speed control was not a core objective.

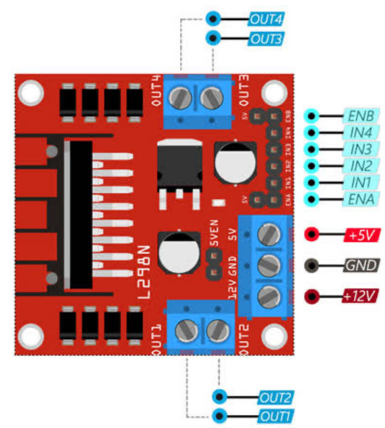

Fig. 3: Pin diagram of the L298N dual H-bridge motor driver used for motor control [3].

The inverted logic from the IR modules ensures that when an obstacle is detected, the corresponding motor reverses its direction, facilitating simple yet effective obstacle avoidance.

E. Integrated Circuit Overview

The complete circuit schematic, including the IR detection modules, RTL-based NOT gates, and motor driver interconnections, is presented in Fig. 4. This integrated design forms the basis for a microcontroller-free robotic navigation system. The design philosophy emphasizes a logic gate-driven control scheme using discrete components, which is pedagogically valuable and cost-effective.

F. Decision-Making Logic for Obstacle Avoidance

The robot's navigation behavior is governed by the binary outputs of its two infrared (IR) obstacle detection sensors (IR1

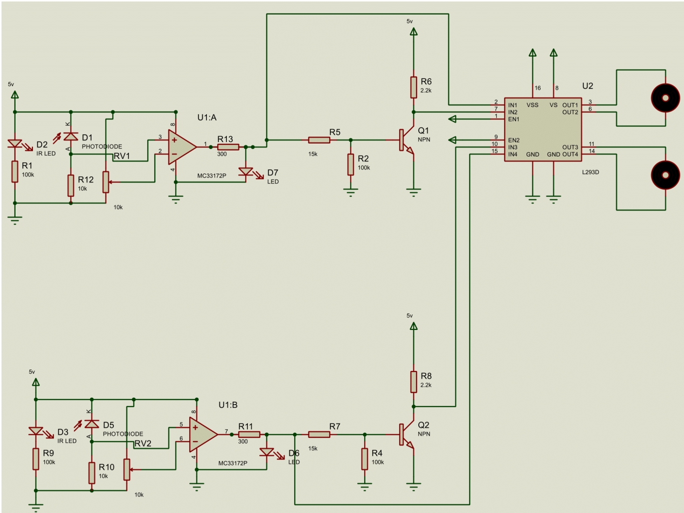

Fig. 4: Complete integrated circuit diagram of the obstacle avoidance robot.

and IR2), each monitoring one side of the forward path. These sensors produce voltage-level signals that are inverted using discrete RTL-based NOT gates, resulting in logic signals that are fed into the L298N motor driver for motor control.

Table I presents the complete truth table describing how the robot responds under different combinations of obstacle detection. In this table:

- OB1 and OB2 are binary obstacle detection flags for IR1 and IR2, respectively, where `0` denotes no obstacle and `1` denotes obstacle presence.
- IR1 and IR2 indicate the analog voltage output from the sensors.
- NOT1 and NOT2 represent the digital output from the RTL NOT gates.
- M1 and M2 denote the motor directions for the left and right motors, respectively (`FW` = forward, `BW` = backward).

The robot's decision-making logic is described as follows:

- **Case 1: OB1 = 0, OB2 = 0**
  Neither sensor detects an obstacle. Both IR1 and IR2 output approximately 3.3V, which results in logic LOW outputs from the NOT gates. This causes both motors to rotate forward (`M1 = FW, M2 = FW`), and the robot moves straight ahead.
- **Case 2: OB1 = 0, OB2 = 1**
  An obstacle is detected by IR2 (right side), while IR1 detects none. IR2 outputs a low voltage, which is inverted by NOT2 to logic HIGH. As a result, the left motor moves forward and the right motor moves backward (`M1 = FW, M2 = BW`), causing the robot to turn left to avoid the obstacle.
- **Case 3: OB1 = 1, OB2 = 0**
  An obstacle is detected by IR1 (left side), while IR2 detects none. The low voltage output from IR1 is inverted to HIGH by NOT1. In this condition, the left motor moves backward and the right motor moves forward (`M1 = BW, M2 = FW`), prompting a right turn to avoid the obstacle.
- **Case 4: OB1 = 1, OB2 = 1**
  Both sensors detect obstacles. The outputs from IR1 and IR2 are low, leading both NOT gates to produce logic HIGH. This results in both motors reversing (`M1 = BW, M2 = BW`), causing the robot to retreat backward from the obstacle-laden area.

This behavior illustrates a simple yet effective reactive control scheme, where the robot dynamically adjusts its movement using Boolean logic derived from sensor input. The absence of a microcontroller in this configuration highlights the feasibility of real-time decision-making using low-cost discrete hardware components.

TABLE I: Robot Behavior Based on Sensor Input

OB1	OB2	IR1	IR2	NOT1	NOT2	M1	M2
0	0	3.3V	3.3V	Low	Low	FW	FW
0	1	3.3V	400mV	Low	High	FW	BW
1	0	400mV	3.3V	High	Low	BW	FW
1	1	400mV	400mV	High	High	BW	BW

III. Conclusion

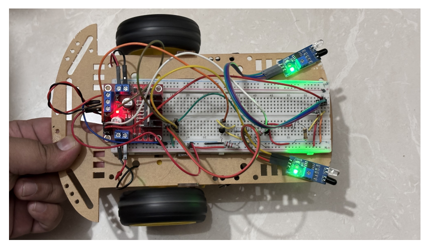

Fig. 5: Schematic of the IR sensor module showing internal op-amp-based comparator and external BJT-based NOT gate logic.

The obstacle avoider robot demonstrates how simple electronic components and logic design can be used to implement intelligent behavior. The use of discrete components to build NOT gates gives insight into the working principles of digital logic without microcontrollers. This system successfully avoids obstacles by reacting to sensor inputs and dynamically adjusting motor directions, making it ideal for educational purposes and early robotics experimentation.

Acknowledgment

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